

Quantitative Trading With Machine Learning

Problem #43 — One-/Two-Page Project Write-up

Target asset: VWAGY (Volkswagen AG) | Academic ML & Quantitative Finance Experiment

1. Problem Statement

The project investigates whether machine-learning models can use historical Volkswagen market information and publicly available supply-chain proxy information to predict future VWAGY stock returns. The goal is not to claim guaranteed profitability, but to construct a leakage-aware quantitative forecasting pipeline and evaluate whether the resulting predictions contain useful out-of-sample signal.

2. Dataset Details

Daily market data were collected from Yahoo Finance for VWAGY and nine public companies used as practical supply-chain proxies: Continental (CON.DE), Infineon (IFX.DE), HELLA (HLE.DE), BASF (BAS.DE), BorgWarner (BWA), Magna (MGA), Lear (LEA), Aptiv (APTV), and TSMC (TSM). These proxies are not claimed to be the exact supplier list from the reference study. The configured market-data period is 2015–2025.

The final engineered dataset contains 276 features: 24 VWAGY technical features and 252 supply-chain features (9 proxies $\times$ 28 lagged daily returns). Forward-return targets were constructed for 1, 5, 10, 15, 20, and 25 trading days. Short data gaps were forward-filled for at most five days, while incomplete rows after feature and target construction were removed.

3. Approach

The experiment follows a paper-inspired rolling time-series methodology rather than a random train/test split. For each monthly rolling window, approximately five years of historical data are used for training, the following year is used for validation, and the next month is kept as unseen out-of-sample test data. The training period expands as the evaluation moves forward.

Component	Implementation
Models	Elastic Net, Decision Tree, XGBoost, LightGBM
Horizons	1, 5, 10, 15, 20, 25 trading days
Selection	Lowest validation RMSE; test data not used for selection
OOS evaluation	55 rolling windows; 1,150 out-of-sample observations
Selected horizon	1-day horizon selected in all 55 windows

4. Implementation Overview

- **Data collection:** Download and align VWAGY and proxy-company daily market data.
- **Feature engineering:** Construct VWAGY return, volatility, absolute-return and momentum features over 3–60 day windows, plus 28 historical return lags for every proxy.
- **Target construction:** Calculate future VWAGY returns at multiple forecast horizons.
- **Model selection:** Evaluate all four models across all six horizons on each validation window and select the lowest-RMSE configuration.
- **Out-of-sample prediction:** Retrain the selected configuration on training + validation data and predict the next unseen month.

5. Results and Evaluation

The current run produced 55 rolling windows and 1,150 out-of-sample observations. The 1-day forecast horizon was selected in every window, while the preferred model varied across market periods. Elastic Net was selected 27 times, XGBoost 18 times, Decision Tree 8 times, and LightGBM 2 times. This variation indicates that model performance is regime-dependent rather than uniformly dominated by a single algorithm.

Several individual test windows produced positive R^2 values (including windows 5, 11, 16, 17, 18, 47, 48 and 51), but these successes were intermittent. The complete out-of-sample evaluation is therefore more important than selecting individual profitable or high- R^2 periods. The notebook calculates aggregate RMSE, MAE, R^2 and prediction correlation directly from the full OOS prediction set.

6. Trading Strategy

Predictions are converted into positions using four prediction thresholds: 0.00%, 0.25%, 0.50% and 1.00%. A prediction above the positive threshold generates a LONG position, a prediction below the negative threshold generates a SHORT position, and predictions inside the threshold produce a NEUTRAL position. The strategy is compared with a VWAGY Buy-and-Hold benchmark.

The backtest reports total return, annualized return, annualized volatility, Sharpe ratio, maximum drawdown, win rate and active trading days. It is an academic backtest and does not fully model bid-ask spreads, slippage, market impact, liquidity constraints, borrowing costs or real order execution.

7. Leakage Prevention

- Chronological rolling evaluation is used instead of random splitting.
- Features use only information available at the prediction date; future prices are used only to construct targets.
- Model and forecast-horizon selection use validation data only.
- The selected model is retrained before predicting the next unseen test month.
- Elastic Net scaling is performed inside a training-data pipeline.

8. Conclusions

The project successfully implements an end-to-end quantitative machine-learning workflow for VWAGY return forecasting using both market features and supply-chain proxies. The experiment demonstrates how to construct forward-return targets, engineer lagged supplier information, compare different regression models, perform rolling out-of-sample evaluation, and translate predictions into trading signals.

The results also show why financial prediction must be evaluated across genuinely unseen periods. Although some rolling windows produced positive test R^2 , the preferred model changed over time and no single model consistently dominated. The experiment therefore does not establish a stable predictive advantage from the tested feature set. The negative or weak aggregate predictive behaviour, if observed in the notebook's final aggregate metrics, should be interpreted as evidence that this particular information set and modelling configuration did not reliably outperform a simple mean-return baseline across the full evaluation period—not as evidence that machine learning cannot be useful for quantitative finance.

Future improvements: expand the proxy universe, add macroeconomic variables, perform stronger walk-forward or nested validation, tune hyperparameters systematically, test statistical significance and feature stability, and introduce more realistic transaction-cost and execution modelling.

Note: This project is an academic research experiment and is not financial advice or a recommendation to buy, sell, or short any security.